

CAPIBARA Collaboration: the CRD Instrument and the COSMOS Program



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Educational Activities -
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CAPIBARA Collaboration

The CAPIBARA Collaboration

The **Collaboration for the Analysis of Photonic and Ionic Bursts and Radiation (CAPIBARA)** is an international group of undergraduate and high school students designing and building space-based detectors. We operate to projects in parallel: a pathfinder Cosmic Ray Detector (CRD) and the long-term COSMOS program.

CAPIBARA is **fully run by students** providing them with hands-on, real-life engagement and experience in these fields. Beyond our projects we want to leverage accessible CubeSat technology to demonstrate what students are capable.



Cosmic Ray Detector (CRD)

Goal

Detect and study the flux and energy spectrum of primary cosmic rays (CRs): charged, high-energy particles with relativistic speeds from outer space or the Sun.

Limitation

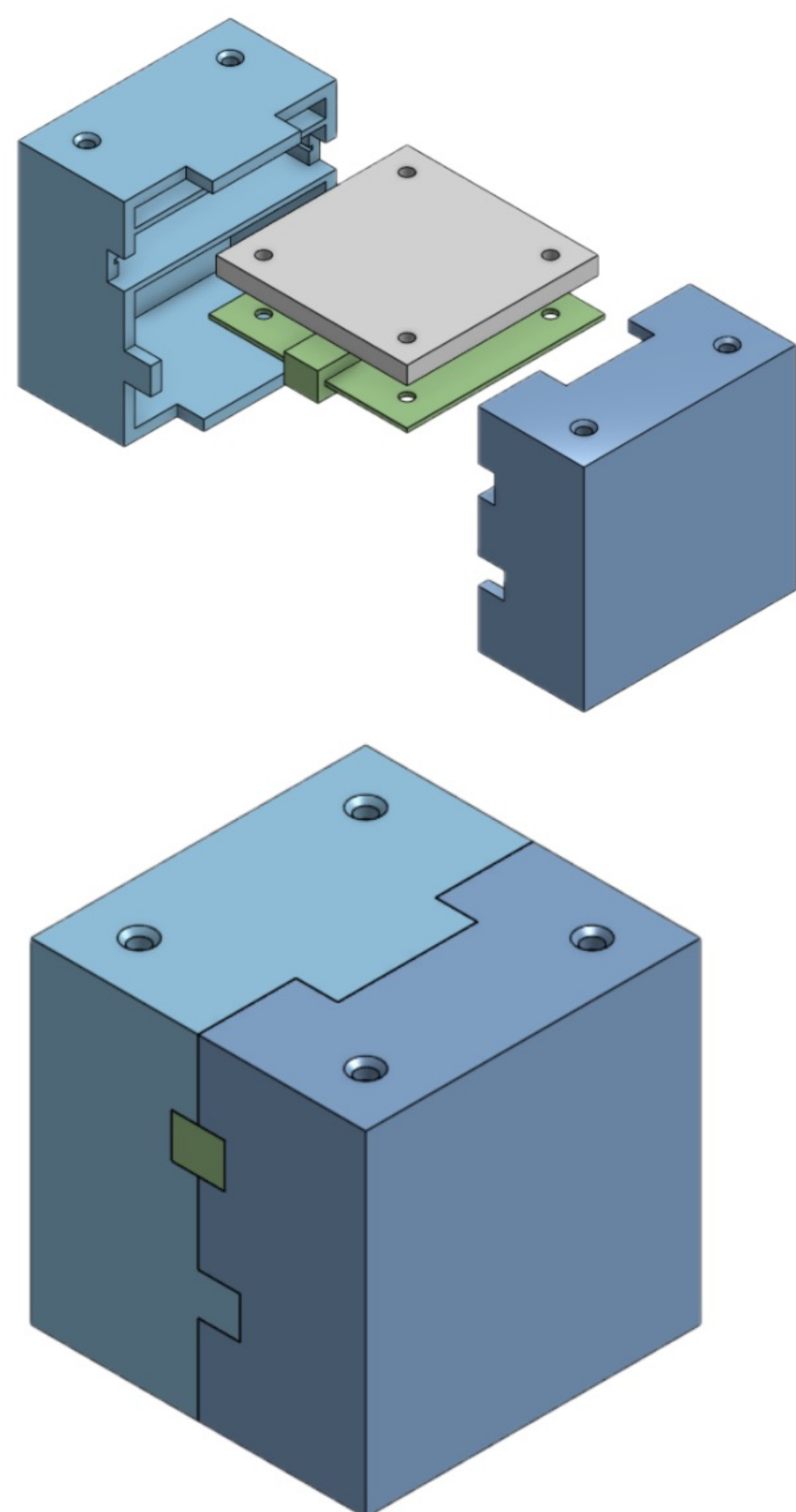
These particles get blocked by Earth's atmosphere, therefore we aim to insert our CR detector into orbit. An ambitious goal that we plan to achieve thanks to the partnership with OBA Space, which will deploy our instrument as a payload on the OAB FARADAY satellite.

Detector

1. TOP: plastic scintillator, excited by passes of charged particles, connected to the silicon photomultiplier (SiPM) which transforms the emitted photons into electric signals.
2. CENTRAL COMPARTMENT: will hold the electrical circuit.
3. BOTTOM COMPARTMENT: it will keep the batteries and the memory storage needed.

Next Steps

Currently we are working on the detector 3D model with guidance and technical support from PLD Space's SPARK program (pldspace.com) alongside OBA Space (oba.space).



Constellation Of Student-Made Observing CubeSats (COSMOS)

Goal (CAPIBARA's long-term strategic vision)

To contribute to the coming era of multi-messenger astronomy by detecting and rapidly localising high-energy transient events, electromagnetic counterparts to GW sources.

Scientific motivation

Next-generation gravitational wave observatories, will detect hundreds to thousands of events annually, but complete studies of these phenomena and their sources require multi-messenger observations. High-energy transients like GRBs provide crucial early detection, as brief Gamma-ray/X-ray emissions (prompt) are followed by emissions in longer wavelengths (afterglow). **Current capabilities face a coverage gap!** Our aim is to constantly monitor the sky and provide fast alerts assisting the job of flagship observatories.

Technical Approach

By building on existing heritage (GRBAlpha, Hermes, EIRSAT-1, etc.) rather than developing novel detectors, we reduce risk and focus innovation on constellation coordination, autonomous transient recognition, and time-delay source localisation. The experience and observations obtained with COSMOS-Twin will serve us well for this purpose.

Our advisory board will provide mentorship and advice to approach these challenges, making COSMOS a valuable training group for students. This advisory board is currently in the process of being established.

Phased Mission Architecture

Phase 0 (2024-2026)

field status study, mission concept development
create collaboration, build documentation,
advisory board, culminating in SSEA 2026.

Phase 1: COSMOS TWIN (2026-2032)

Deployment of two 3-6U CubeSats to
demonstrate time-delay localisation (using
GPS-synchronised clocks). We expect to be
able to participate in a public or commercial
rideshare launch program like ESA's FYS!.

Phase 2: COSMOS-Net (2031-2040)

Scales number of satellites in constellation,
enabling true triangulation for arcminute-
level localisation and more monitoring.
Collaboration expands to multiple
universities.